CPU Scheduling

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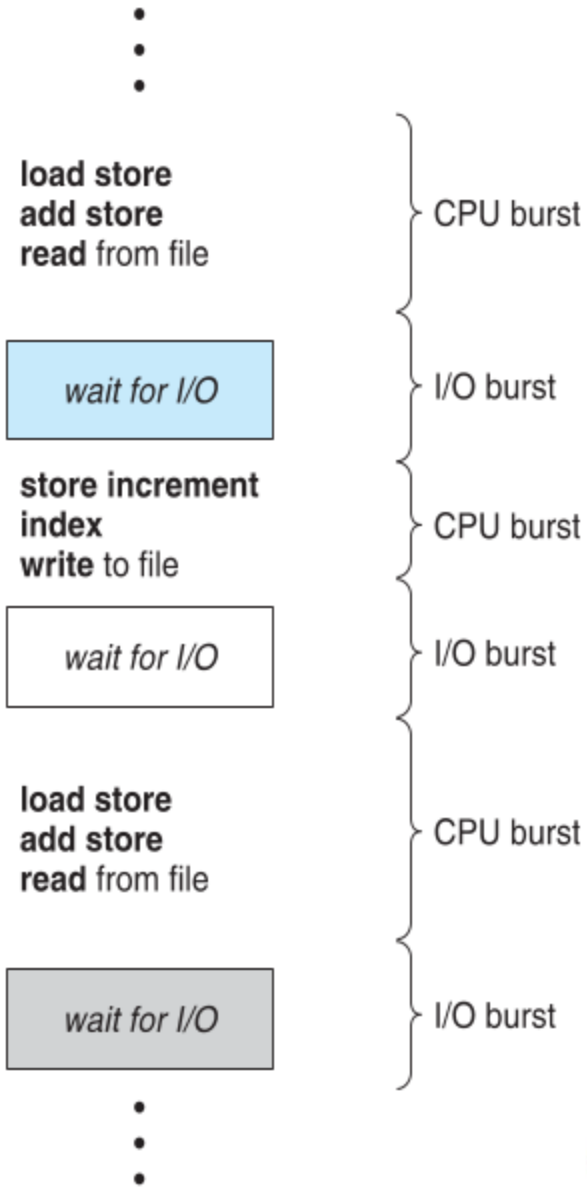
https://github.com/yoosofan/slide/blob/main/os/cpu.rst

CPU Burst / Service Time

- cpu utilization by multiprogramming
- cpu - I/O cycle of a process
- cpu : burst time, service time
- I/O : or other blocking events like memory request

Process behaviour

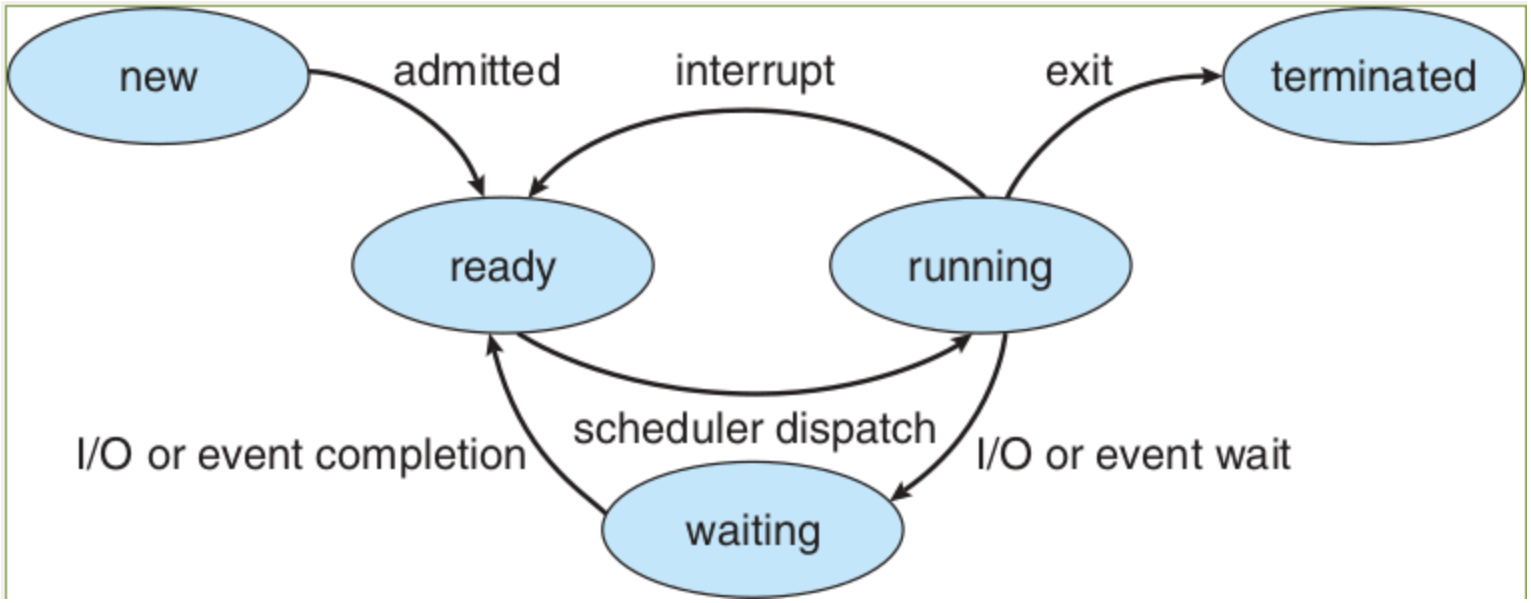
1. cpu bound
2. I/O bound



CPU Schedular

Short Term Schedular

- ready queue
- Dispatcher



Time Unit Concept

- Millisecond
- Nanosecond
- ?

Scheduling type

- nonpreemptive
- preemptive

Processes Table

process	service(Burst) time
P_0	3
P_1	2
P_2	1
P_3	2

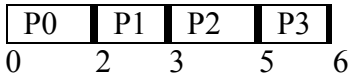
prs	st
P_0	3
P_1	2
P_2	1
P_3	2

process	service time	arrival time
P_0	3	0
P_1	2	0
P_2	1	3
P_3	2	5

pcs	st	at
P_0	3	0
P_1	2	0
P_2	1	3
P_3	2	5

First-Come, First-Served (FCFS)

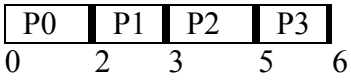
process	service time	arrival time
P_0	2	0
P_1	1	0
P_2	2	3
P_3	1	4



- Gantt Chart
- t = 0: ready queue(q) = [P_0, P_1]
- t = 2: q = [P_1]
- t = 3: q = [P_2]
- t = 5: q = [P_3]

Average Waiting Time

process	service time	arrival time
P_0	2	0
P_1	1	0
P_2	2	3
P_3	1	4



- P_0 waiting time: 0
- P_1 waiting time: 2
- P_2 waiting time: (3-3) = 0
- P_3 waiting time: (5-4) = 1

- Average Waiting Time
- $\frac{0+2+0+1}{4} = \frac{3}{4} = 0.75$

FCFS - Convoy effect

process	service time	arrival time
P_0	4	0
P_1	6	0
P_2	1	3
P_3	3	4

P0	P1	P2	P3

0 4 10 11 14

- Average Waiting Time
- $\frac{0+(4-0)+(10-3)+(11-4)}{4} = \frac{18}{4} = 4\frac{2}{4} = 4.5$

Rearange

P0	P2	P3	P1

0 4 5 8 14

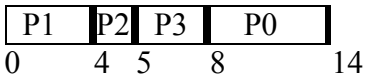
- Average Waiting Time
- $\frac{0+(4-3)+(5-4)+8}{4} = \frac{10}{4} = 2\frac{2}{4} = 2.5$

- Average Waiting Time 1: 4.5
- Average Waiting Time 2: 2.5

SJF or SPN $\frac{1}{s}$

process	service time	arrival time
P_0	6	0
P_1	4	0
P_2	1	3
P_3	3	4

- Shortest Job First (SJF)
- Shortest Process Next (SPN)

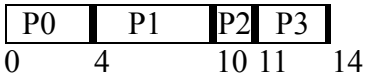


- *Average Waiting Time*
- $\frac{0+(4-3)+(5-4)+8}{4} = \frac{10}{4} = 2\frac{2}{4} = 2.5$

- Starvation
- Nonpreemptive
- formula : 1 / (service time)

Shortest Remaining Time(SRT), preemptive SJF

process	service time	arrival time
P_0	4	0
P_1	6	0
P_2	1	1
P_3	3	2



- Average Waiting Time
- $\frac{0+(4-0)+(10-1)+(11-2)}{4} = \frac{22}{4} = 5\frac{2}{4} = 5.5$

P0	P2	P0	P3	P1

0 1 2 5 8 14

- Average Waiting Time
- $\frac{(0+(2-1))+(8-0)+(1-1)+(5-2)}{4} = \frac{12}{4} = 3$

Hieghest Response Ratio Rate Next (HRRN) $\frac{w+s}{s}$

process	service time	arrival time
p_0	5	0
p_1	3	1
p_2	4	2
p_3	2	6

t = 0 | P_0 | (0 – 5) queue : P1, P2, P3

t = 5 | P_0 | P_1 | (0 – 5 – 8) queue : P2 P3

1. P2: $\frac{(8-2)+4}{4} = \frac{6+4}{4} = \frac{10}{4}$
2. P3: $\frac{(8-6)+2}{2} = \frac{2+2}{2} = \frac{4}{2} = \frac{8}{4}$

t = 8 | P_0 | P_1 | P_2 | (0 – 5 – 8 – 12) queue : P3

HRRN | P_0 | P_1 | P_2 | P_3 | (0 – 5 – 8 – 12 – 14)

SJF | P_0 | P_1 | P_3 | P_2 | (0 – 5 – 8 – 10 – 14)

Average Waiting Time

HRRN: $\frac{0+(5-1)+(8-2)+(12-6)}{4} = \frac{16}{4} = 4$

SJF: $\frac{0+(5-1)+(8-6)+(10-2)}{4} = \frac{14}{4} = \frac{7}{2} = 3.5$

Estimating Service Time(I)

1. $$\tau_n = \frac{t_0 + t_1 + t_2 + \ldots + t_{n-1}}{n}$$

2. $$n * \tau_n = t_0 + t_1 + t_2 + \ldots + t_{n-1}$$

3. $$\tau_{n+1} = \frac{t_0 + t_1 + t_2 + \ldots + t_{n-1} + t_n}{n + 1}$$

4. $$= \frac{t_0 + t_1 + t_2 + \ldots + t_{n-1}}{n + 1} + \frac{t_n}{n + 1}$$

5. $$\tau_{n+1} = \frac{n * \tau_n}{n + 1} + \frac{t_n}{n + 1}$$

6. $$\tau_{n+1} = \frac{n}{n + 1} * \tau_n + \frac{1}{n + 1} * t_n$$

Estimating Service Time(II)

1. $$\tau_{n+1} = \frac{n}{n+1} * \tau_n + \frac{1}{n+1} * t_n$$

2. $$\tau_{n+1} = \frac{n+1-1}{n+1} * \tau_n + \frac{1}{n+1} * t_n$$

3. $$\tau_{n+1} = \left(\frac{n+1}{n+1} - \frac{1}{n+1}\right) * \tau_n + \frac{1}{n+1} * t_n$$

4. $$\tau_{n+1} = \left(1 - \frac{1}{n+1}\right) * \tau_n + \frac{1}{n+1} * t_n$$

5. $$\alpha = \frac{1}{n+1}$$

$$\tau_{n+1} = (1-\alpha) * \tau_n + \alpha * t_n$$

Estimating Service Time(III)

1. $$\alpha = \frac{1}{n+1}, \quad \tau_{n+1} = (1-\alpha) * \tau_n + \alpha * t_n$$

2. $$t_n = \text{actual length of } n^{th} \text{ service time}$$

3. $$\tau_{n+1} = \text{predicted value for the next service time}$$

4. $$0 \leqslant \alpha \leqslant 1, \quad \tau_{n+1} = (1-\alpha) * \tau_n + \alpha * t_n$$

5. $$\alpha \rightarrow 0$$

Round Robin (RR , quantum) I

process	service time	arrival time
p_0	5	0
p_1	3	1
p_2	4	2
p_3	2	6

time quantum or q = 2 , Queue (Q): Empty

t = 0 , Q: P0

P_0

0 2

t=2, Q: P1(3), P2(4), P0(3)

P_0	P_1

0 2 4

t=4, Q:P2(4), P0(3), P1(1)

P_0	P_1	P_2

0 2 4 6

t=6, Q: P0(3), P1(1), P3(2), P2(2)

P_0	P_1	P_2	P_0

0 2 4 6 8

t=8, Q: P1(1), P3(2), P2(2), P0(1)

P_0	P_1	P_2	P_0	P_1

0 2 4 6 8 9

t=9, Q: P3(2), P2(2), P0(1)

P_0	P_1	P_2	P_0	P_1	P_3

0 2 4 6 8 9 11

t=11, Q: P2(2), P0(1)

P_0	P_1	P_2	P_0	P_1	p_3	p_2

0 2 4 6 8 9 11 13

t=13, Q: P0(1)

P_0	P_1	P_2	P_0	P_1	p_3	p_2	p_0

0 2 4 6 8 9 11 13 14

Round Robin (RR) II

process	service time	arrival time
p_0	5	0
p_1	3	1
p_2	4	2
p_3	2	6

t=11, Q: P2(2), P0(1)

- P_0 P_1 P_2 P_0 P_1 p_3 p_2
  0 2 4 6 8 9 11 13

t=13, Q: P0(1)

- P_0 P_1 P_2 P_0 P_1 p_3 p_2 p_0
  0 2 4 6 8 9 11 13 14

Average Waiting Time

$$\frac{[0+(6-2)+(13-8)]+[(2-1)+(8-4)]+[(4-2)+(11-6)]+[9-6]}{4}$$

$$= \frac{9+5+7+3}{4} = \frac{24}{4} = 6$$

Priority

1. Internal
2. External

- smallest integer ≡ highest priority
- largest integer ≡ highest priority

1. preemptive (absolute)
2. non-preemptive (relative)

- Starvation
  - Aging

nice [-20 , 19]

```
root@computer-name:~# nice --5 geany
root@computer-name:~# ps -l
root@computer-name:~# top
user@computer-name:~# nice -n 8 geany
```

renice

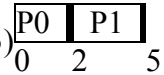
```
user@computer-name:~# renice 10 -p 19862
user@computer-name:~# renice -n 15 -p 19862
```

Relative Priority

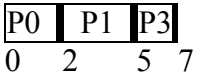
process	service time	arrival time	Priority
P0	2	0	4
P1	3	1	3
P2	1	2	3
P3	2	5	1

t=0, Q: P0(2,4)

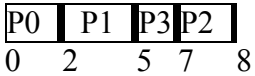
t=2, Q: P1(3,3), P2(1,3)



t=5, Q: P2(1,3), P3(2,1)



t=6, Q: P2(1,3)

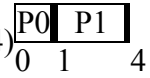


Absolute Priority

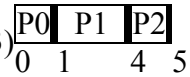
process	service time	arrival time	Priority
P0	2	0	4
P1	3	1	3
P2	1	2	3
P3	2	5	1

t=0, Q: P0(2,4)

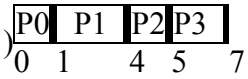
t=1, Q: P1(3,3), P0(1,4)



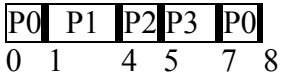
t=4, Q: P0(1,4), P2(1,3)



t=5, Q: P0(1,4), P3(2,1)



t=7, Q: P0(1,4)



Priority Round Robin

process	service time	arrival time	Priority
P0	2	0	4
P1	3	1	3
P2	1	2	2
P3	2	5	1

t=0, Q: P0(2)

P0

0 1

t=1, Q: P0(1,4), P1(3,3)

P0	P1

0 1 2

t=2, Q: P0(1,4), P1(2,3), P2(1,2)

P0	P1	P2

0 1 2 3

t=3, Q: P0(1,4), P1(2,3)

P0	P1	P2	P1

0 1 2 3 5

t=5, Q: P0(1,4), P3(2,1)

P0	P1	P2	P1	P3

0 1 2 3 5 7

t=7, Q: P0(1,4)

P0	P1	P2	P1	P3	P0

0 1 2 3 5 7 8

Multilevel Queue

process	service time	arrival time
p_0	5	0
p_1	3	1
p_2	4	2
p_3	2	6

t=11, Q: P2(2), P0(1)

- P_0 P_1 P_2 P_0 P_1 p_3 p_2
  0 2 4 6 8 9 11 13

t=13, Q: P0(1)

- P_0 P_1 P_2 P_0 P_1 p_3 p_2 p_0
  0 2 4 6 8 9 11 13 14

Average Waiting Time

$$\frac{(0+(6-2)+(13-8))+((2-1)+(8-4))+((4-2)+(11-6))+(9-6)}{4}$$

$$= \frac{9+5+7+3}{4} = \frac{24}{4} = 6$$

Multilevel Queue Feedback (MLQF)

process	service time	arrival time
P_0	5	0
P_1	3	1
P_2	4	2
P_3	2	6

t=11, Q: P2(2), P0(1)

- P_0 P_1 P_2 P_0 P_1 p_3 p_2
  0 2 4 6 8 9 11 13

t=13, Q: P0(1)

- P_0 P_1 P_2 P_0 P_1 p_3 p_2 p_0
  0 2 4 6 8 9 11 13 14

Average Waiting Time

$$\frac{(0+(6-2)+(13-8))+((2-1)+(8-4))+((4-2)+(11-6))+(9-6)}{4}$$

$$= \frac{9+5+7+3}{4} = \frac{24}{4} = 6$$

Scheduling Criteria

1. *Waiting time*: Min
2. *CPU utilization*: Max
3. *Throughput*: Max
4. *Turnaround time*: Min
5. *Response time*: Min
    1. Optimainzation
    2. Operationg Systems

- *Fairness*
- *No starvation*

END